

BHAKTI LAMBATE

VLSI / RTL DESIGN | COMPUTER ARCHITECTURE | SYSTEMVERILOG | VERIFICATION

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SUMMARY

VLSI Design & Technology undergraduate with 8.96 CGPA and Honors in AI/ML, specializing in high-performance RTL design, computer architecture, and ASIC implementation. Architected RISC-V CPU pipelines, parameterized systolic AI/graphics accelerators, and cache-coherent NoC architectures in SystemVerilog, spanning microarchitecture, datapath/control design, verification, synthesis, timing, and PPA optimization. Experienced in translating complex architectural concepts into scalable, robust, and performance-driven hardware through rigorous functional and formal verification and RTL-to-GDSII implementation.

SKILLS & TECHNOLOGIES

HDL & RTL	Verilog, SystemVerilog, RTL & SoC Design, Microarchitecture, Computer Architecture, SVA, Clock Domain Crossing (CDC)
Architecture	RISC-V, CPU Pipelining, Hazard Handling, Cache Coherence, NoC, Memory Systems
Verification	UVM, Functional/Formal Verification, cocotb, Verilator, Icarus Verilog, Self-Checking Testbenches
ASIC/EDA	Yosys, OpenLane, OpenROAD, Synopsys Design Compiler, PrimeTime, Vivado, STA
Programming	Python, C, C++, TCL, Bash, Perl, Git, Linux

EDUCATION

- Bachelor of Engineering - Electronics Engineering (VLSI Design & Technology)** July 2023- May 2027
Honors- Artificial Intelligence & Machine Learning
Current CGPA- 8.96

PROJECTS

Parameterized Systolic Accelerator for AI/Graphics Workloads | Jun 2026- Aug 2026

- Architected a parameterized $N \times N$ systolic-array accelerator with pipelined MAC Processing Elements (PEs), local registers, and data-skew buffers for high-throughput parallel computation and efficient data reuse.
- Built a self-checking, cycle-accurate verification environment using SystemVerilog, cocotb, Python golden models, Verilator/Icarus Verilog, SVA, and Symbyosys, covering functional, randomized, lint, and assertion-based validation.
- Executed an open-source RTL-to-GDSII ASIC flow using Yosys, OpenLane, and OpenROAD, covering logic synthesis, PPA optimization, timing analysis, floorplanning, placement, routing, and physical-design evaluation.

5-Stage Pipelined RISC-V Processor Core | Jan 2026 – Apr 2026

- Designed and implemented a 32-bit RV32I 5-stage pipelined CPU microarchitecture (IF/ID/EX/MEM/WB) with ALU, register file, control unit, immediate generation, and instruction/data memory interfaces.
- Implemented hazard detection, data forwarding, pipeline stalls, branch/control-flow handling, and memory operations using self-checking RTL testbenches and RISC-V ISA-based functional verification.
- Synthesized RTL using Yosys and evaluated timing, area, and PPA characteristics, validating the processor through an open-source RTL-to-synthesis flow.

Cache-Coherent Network-on-Chip (NoC) for Multicore RISC-V SoC | Nov 2025- Dec 2025

- Architected a scalable 2D-mesh NoC microarchitecture with packet-based communication, XY routing, input buffering, arbitration, flow control, and configurable network parameters for multicore RISC-V SoC interconnects.
- Implemented and verified cache-coherence and memory-transaction logic using SystemVerilog, cocotb, Verilator, and SVA/formal verification, validating packet delivery, ordering, arbitration, deadlock-free routing, and corner-case traffic.
- Synthesized RTL using Yosys and evaluated latency, throughput, congestion, area, timing, and PPA using an open-source RTL-to-ASIC flow with OpenROAD for physical-design analysis.

ACHIEVEMENTS

- Smart India Hackathon 2025 Finalist:** Selected as a Finalist in Smart India Hackathon 2025, demonstrating problem-solving, rapid prototyping and collaborative technical development.
- Fellowship- International VLSI Design Conference 2026:** Participated in technical sessions covering semiconductor design and RISC-V architectures. Engaged with industry professionals and EDA leaders on emerging trends in the semiconductor ecosystem.

EXPERIENCE

System Testing Intern | National Test House | Government of India Ministry of Consumer Affairs, Food & Public Distribution

Dec 2025- Jan 2026

- Executed system-level testing, verification, and validation for mission-critical platforms, including a Laboratory Information Management System (LIMS) and supported the quality assurance lifecycle, identifying functional issues and validating system behavior against defined requirements.
- Worked with structured testing and validation processes to ensure reliable system operation. Ensured compliance with applicable government regulatory and quality standards and supported rigorous QA protocols.